ModuleDict#

https://pytorch.org/docs/stable/generated/torch.nn.ModuleDict.html#torch.nn.

Description:

Holds submodules in a dictionary. ModuleDict can be indexed like a regular Python dictionary, but modules it contains are properly registered, and will be visible by all Module methods. ModuleDict is an ordered dictionary that respects the order of insertion, and in update(), the order of the merged OrderedDict, dict (started from Python 3.6) or another ModuleDict (the argument to update()).

Function Signature

```
class torch.nn.ModuleDict(modules=None)[source]#
Parameters
clear()[source]#
items()[source]#
Return type
```

Returns:

ValuesView[Module]

Code Examples

```
class MyModule(nn.Module):
    def __init__(self) -> None:
        super().__init__()
        self.choices = nn.ModuleDict(
            {"conv": nn.Conv2d(10, 10, 3), "pool":
nn.MaxPool2d(3)}
        )
        self.activations = nn.ModuleDict(
            [["lrelu", nn.LeakyReLU()], ["prelu", nn.PReLU()]]
        )

    def forward(self, x, choice, act):
        x = self.choices[choice](x)
        x = self.activations[act](x)
        return x
```

Notes & Warnings

NOTE Note If modules is an OrderedDict, a ModuleDict, or an iterable of key-value pairs, the order of new elements in it is preserved.

Detailed Documentation

Documentation

Holds submodules in a dictionary.

ModuleDict can be indexed like a regular Python dictionary, but modules it contains are properly registered, and will be visible by all Module methods.

ModuleDict is an ordered dictionary that respects

the order of insertion, and

in update(), the order of the merged OrderedDict, dict (started from Python 3.6) or another ModuleDict (the argument to update()).

Note that update() with other unordered mapping types (e.g., Python's plain dict before Python version 3.6) does not preserve the order of the merged mapping.

modules (iterable, optional) – a mapping (dictionary) of (string: module) or an iterable of key-value pairs of type (string, module)

Remove all items from the ModuleDict.

Return an iterable of the ModuleDict key/value pairs.

ItemsView[str, Module]

Return an iterable of the ModuleDict keys.

Remove key from the ModuleDict and return its module.

key (str) – key to pop from the ModuleDict

Update the ModuleDict with key-value pairs from a mapping, overwriting existing keys.

If modules is an OrderedDict, a ModuleDict, or an iterable of key-value pairs, the order of new elements in it is preserved.

modules (iterable) – a mapping (dictionary) from string to Module, or an iterable of key-value pairs of type (string, Module)

Return an iterable of the ModuleDict values.
